EXCELLENCE

CLASS 11 - 2026-27

Physics | Lecture - 11

Basic Maths and Calculus

Higher Derivatives, Chain Rule and Partial Differentiation

By - Excellence Programme

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Topics to be covered

01 Product and quotient rule continuation

02 5-minute differentiation recap

03 Higher-order derivatives

04 Chain rule: function inside a function

05 S.H.M. application

06 Partial differentiation

07 Parametric and reciprocal derivatives

Product and quotient rules

PRODUCT

$$d(AB)/dx = A\,dB/dx + B\,dA/dx$$

Differentiate one factor at a time.

QUOTIENT

$$d(A/B)/dx = (B\,dA/dx - A\,dB/dx)/B^2$$

Keep the denominator squared.

For a quotient, remember: denominator × derivative of numerator minus numerator × derivative of denominator.

Differentiate a simple polynomial

For $y = 4x^3-5x^2+2x-7$, find dy/dx.

Solution: Differentiate a simple polynomial

Step 1: $d(4x^3)/dx = 12x^2$

Step 2: $d(-5x^2)/dx = -10x$

Step 3: $d(2x)/dx = 2$ and $d(-7)/dx = 0$

$dy/dx = 12x^2-10x+2.$

Rough work / extra work

Product-rule practice

Differentiate $y = \tan x \cdot \log_e x$.

Solution: Product-rule practice

Step 1: Use $d(AB)/dx = A\,dB/dx + B\,dA/dx$

Step 2: $d(\tan x)/dx = \sec^2 x$ and $d(\log_e x)/dx = 1/x$

$dy/dx = \tan x/x + \log_e x \cdot \sec^2 x$.

Stationary-point practice

If $y = A/x^3 - B/x^4$, find x for which $dy/dx = 0$. Assume A and B are non-zero constants.

Solution: Stationary-point practice

Step 1: Write $y = Ax^{-3} - Bx^{-4}$

Step 2: $dy/dx = -3Ax^{-4} + 4Bx^{-5}$

Step 3: Set zero and multiply by x^5: $-3Ax + 4B = 0$

$x = 4B/(3A)$.

Quotient-rule practice: exponential

Differentiate $y = e^x/x^2$, where $x \neq 0$.

Solution: Quotient-rule practice: exponential

Step 1: Take $A = e^x$ and $B = x^2$

Step 2: $dy/dx = [x^2e^x - e^x(2x)]/x^4$

Step 3: Factor e^x x from the numerator

$dy/dx = e^x(x-2)/x^3$.

Rough work / extra work

Quotient-rule practice: algebraic

Differentiate $y = (x^2-1)/(x^2+1)$.

Solution: Quotient-rule practice: algebraic

Step 1: Use the quotient rule with $A = x^2-1$ and $B = x^2+1$

Step 2: $dy/dx = [(x^2+1)(2x)-(x^2-1)(2x)]/(x^2+1)^2$

Step 3: The numerator becomes $2x[(x^2+1)-(x^2-1)] = 4x$

$dy/dx = 4x/(x^2+1)^2$.

Mixed trigonometric-algebraic practice

Differentiate $y = \cos x + 2/x^3$, where $x \neq 0$.

Solution: Mixed trigonometric-algebraic practice

Step 1: Write $2/x^3$ as $2x^{-3}$

Step 2: $d(\cos x)/dx = -\sin x$

Step 3: $d(2x^{-3})/dx = -6x^{-4}$

$dy/dx = -\sin x - 6/x^4$.

Rough work / extra work

Simplify before differentiating

Differentiate $y = (\sqrt{x} + 1/\sqrt{x})^2$ for $x > 0$.

Solution: Simplify before differentiating

Step 1: Expand first: $y = x+2+1/x$

Step 2: Write $1/x$ as x^{-1}

Step 3: $dy/dx = 1+0-x^{-2}$

$dy/dx = 1-1/x^2$.

Differentiation formula map

Function	Derivative
C	0
x^n	nx^{n-1}
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
e^x	e^x
$\log_e x$	$1/x$

Use radians for trigonometric derivatives.

Rules used in the last class

constant multiple: d[kf]/dx = kf'

sum or difference: differentiate each term separately

product: d(AB)/dx = A B' + B A'

quotient: d(A/B)/dx = (B A' - A B')/B^2

Before differentiating, simplify powers and fractions whenever possible.

Recap check: slope at a point

For $y = x^3-5x+x^2$, find dy/dx at $x = 1$.

Solution: Recap check: slope at a point

Step 1: $dy/dx = 3x^2-5+2x$

Step 2: At $x = 1$: $dy/dx = 3-5+2$

Slope at $x = 1$ is 0.

Rough work / extra work

Higher-order derivatives

Differentiate y once to get the first derivative.

first derivative: dy/dx

Differentiate the first derivative again to get the second derivative.

second derivative: $d^2y/dx^2 = d/dx(dy/dx)$

Continue in the same way for d^3y/dx^3, d^4y/dx^4 and higher orders.

Meaning of the second derivative

The first derivative measures how y changes with x.

The second derivative measures how that first derivative changes.

position x(t) → velocity v = dx/dt

velocity v(t) → acceleration a = dv/dt = d^2x/dt^2

For a graph, the second derivative helps describe its curvature.

Repeated derivatives of sin x

$y = \sin x$

$\frac{dy}{dx} = \cos x$

$\frac{d^2y}{dx^2} = -\sin x$

$\frac{d^3y}{dx^3} = -\cos x$

$\frac{d^4y}{dx^4} = \sin x$

The derivatives repeat after every four differentiations.

Higher derivatives: mixed function

For $y = 4x^3 + 3\sin x - 4e^x$, find the first four derivatives.

Solution: Higher derivatives: mixed function

Step 1: $dy/dx = 12x^2+3\cos x-4e^x$

Step 2: $d^2y/dx^2 = 24x-3\sin x-4e^x$

Step 3: $d^3y/dx^3 = 24-3\cos x-4e^x$

Step 4: $d^4y/dx^4 = 3\sin x-4e^x$

Rough work / extra work

Higher derivatives: polynomial

For $y = x^5+2x^4+3x^3+4x^2+5x+6$, find derivatives up to d^4y/dx^4.

Solution: Higher derivatives: polynomial

Step 1: $dy/dx = 5x^4+8x^3+9x^2+8x+5$

Step 2: $d^2y/dx^2 = 20x^3+24x^2+18x+8$

Step 3: $d^3y/dx^3 = 60x^2+48x+18$

Step 4: $d^4y/dx^4 = 120x+48$

Chain rule: a function inside a function

Use the chain rule when one function is placed inside another.

$\sin(x^3)$: sine outside, x^3 inside

$e^{\sin x}$: exponential outside, sin x inside

$(2x+3)^5$: fifth power outside, 2x+3 inside

Product rule is for multiplied functions. Chain rule is for nested functions.

Outside–inside rule

Let y = f(u) and u = g(x). Differentiate in two stages.

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$$

Step 1: Differentiate the outer function, keeping the inner expression unchanged.

Step 2: Multiply by the derivative of the inner expression.

$$\frac{d}{dx}[f(g(x))] = f'(g(x))g'(x)$$

Rough work / extra work

Recognising the inner function

$y = \sin(x^3)$

$\frac{dy}{dx} = \cos(x^3) \cdot 3x^2$

$y = \cos(e^x)$

$\frac{dy}{dx} = -\sin(e^x) \cdot e^x$

Outer derivative keeps the inner expression as it is.

outer derivative × inner derivative

Chain rule: trigonometric function

Differentiate $y = \sin(2x^3+4)$.

Solution: Chain rule: trigonometric function

Step 1: Outer derivative: $\cos(2x^3+4)$

Step 2: Inner derivative: $d(2x^3+4)/dx = 6x^2$

$dy/dx = 6x^2\cos(2x^3+4)$.

Chain rule: exponential of x^2

Differentiate $y = e^{x^2}$.

Solution: Chain rule: exponential of x^2

Step 1: Outer function e^u differentiates to e^u

Step 2: Inner function $u = x^2$ has $du/dx = 2x$

$dy/dx = 2x\, e^{x^2}$.

Rough work / extra work

Chain rule: exponential of sin x

Differentiate $y = e^{\sin x}$.

Solution: Chain rule: exponential of sin x

Step 1: Outer derivative: $e^{\sin x}$

Step 2: Inner derivative: d(sin x)/dx = cos x

$dy/dx = e^{\sin x} \cos x$.

Chain rule: power of a function

Differentiate $y = (x^2+2x)^3$.

Solution: Chain rule: power of a function

Step 1: Keep the inner expression: $3(x^2+2x)^2$

Step 2: Differentiate inside: $d(x^2+2x)/dx = 2x+2$

$dy/dx = 3(x^2+2x)^2(2x+2)$.

Chain rule: linear expression

Differentiate $y = (2x+3)^5$.

Solution: Chain rule: linear expression

Step 1: Outer derivative: $5(2x+3)^4$

Step 2: Inner derivative: $d(2x+3)/dx = 2$

$dy/dx = 10(2x+3)^4$.

Rough work / extra work

Chain rule: logarithm

Differentiate $y = \log_e(\sin x)$.

Solution: Chain rule: logarithm

Step 1: For $\log_e u$: derivative = $(1/u)(du/dx)$

Step 2: $u = \sin x$ and $du/dx = \cos x$

$dy/dx = \cos x/\sin x = \cot x$.

Chain rule: sine of a logarithm

Differentiate $y = \sin(\log_e x)$, where $x > 0$.

Solution: Chain rule: sine of a logarithm

Step 1: Outer derivative: $\cos(\log_e x)$

Step 2: Inner derivative: $d(\log_e x)/dx = 1/x$

$dy/dx = \cos(\log_e x)/x.$

Rough work / extra work

Chain rule: square root

Differentiate $y = \sqrt{(x+4)}$, where $x > -4$.

Solution: Chain rule: square root

Step 1: Write $y = (x+4)^{1/2}$

Step 2: $dy/dx = (1/2)(x+4)^{-1/2} \cdot 1$

$dy/dx = 1/[2\sqrt{(x+4)}]$.

Chain rule: square of sine

Differentiate $y = \sin^2 x$.

Solution: Chain rule: square of sine

Step 1: Treat $y = (\sin x)^2$

Step 2: $dy/dx = 2\sin x \cdot \cos x$

$dy/dx = \sin(2x)$.

Nested chain rule

Differentiate $y = e^{\sin(x^3)}$.

Solution: Nested chain rule

Step 1: Differentiate e^u: $e^u \cdot du/dx$

Step 2: For $u = \sin(x^3)$: $du/dx = \cos(x^3) \cdot 3x^2$

$dy/dx = 3x^2\, e^{\sin(x^3)} \cos(x^3)$.

Rough work / extra work

S.H.M. differential equation

For displacement $y = A \sin(\omega t)$, differentiate with respect to time.

$$\mathbf{dy/dt = A\omega\cos(\omega t)}$$

$$\mathbf{d^2y/dt^2 = -A\omega^2\sin(\omega t)}$$

Since $y = A \sin(\omega t)$, replace $A \sin(\omega t)$ by y.

$$\mathbf{d^2y/dt^2 = -\omega^2 y \quad or \quad d^2y/dt^2 + \omega^2 y = 0}$$

Acceleration is opposite to displacement: $a = -\omega^2 y$.

Partial differentiation

Some physical quantities depend on more than one independent variable.

$$V = V(x, y, z)$$

A partial derivative measures change with respect to one variable while the others are held constant.

$$\partial V/\partial x,\ \partial V/\partial y,\ \partial V/\partial z$$

Hold the other variables constant

To find $\partial V/\partial x$, treat y and z as constants.

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If $V = x^2y$, then $\partial V/\partial x = 2xy$

For the same V, $\partial V/\partial y = x^2$

The symbol ∂ is used because only one variable changes at a time.

Partial derivatives of three variables

For $V = x^2y^2z^4$, find $\partial V/\partial x$, $\partial V/\partial y$ and $\partial V/\partial z$.

Solution: Partial derivatives of three variables

Step 1: $\partial V/\partial x = 2xy^2z^4$

Step 2: $\partial V/\partial y = 2x^2yz^4$

Step 3: $\partial V/\partial z = 4x^2y^2z^3$

Differentiate one variable; keep the other factors unchanged.

Rough work / extra work

Parametric differentiation

Sometimes x and y are both written in terms of a third variable t.

$x = f(t), y = g(t)$

Differentiate each equation with respect to t, then divide.

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}, \text{ provided } \frac{dx}{dt} \neq 0$$

Reciprocal derivative

If dy/dx is known and non-zero, reverse it to obtain dx/dy.

$$\frac{dx}{dy} = \frac{1}{\left(\frac{dy}{dx}\right)}$$

This is useful when the question asks for x changing with y instead of y changing with x.

The reciprocal rule needs $\frac{dy}{dx} \neq 0$.

Parametric derivative

If $y = 2t^3+1$ and $x = t^2$, find dy/dx.

Solution: Parametric derivative

Step 1: $dy/dt = 6t^2$

Step 2: $dx/dt = 2t$

Step 3: $dy/dx = (6t^2)/(2t)$

$dy/dx = 3t$, for $t \neq 0$.

Rough work / extra work

Reciprocal derivative

If $y = \sin u$ and $u = 3x$, find dx/dy.

Solution: Reciprocal derivative

Step 1: $dy/du = \cos u$ and $du/dx = 3$

Step 2: $dy/dx = 3\cos u = 3\cos(3x)$

Step 3: Take the reciprocal

$dx/dy = 1/[3\cos(3x)]$, where $\cos(3x) \neq 0$.

Slope using differentiation

Find the slope of $y = 3x^2+5x-4$ at $x = 1$.

Solution: Slope using differentiation

Step 1: $dy/dx = 6x+5$

Step 2: At $x = 1$: slope $= 6(1)+5$

Slope = 11.

Homework: trigonometric chain rule

Differentiate $y = A\cos(kx)$, where A and k are constants.

Solution: Homework: trigonometric chain rule

Step 1: Outer derivative: $-A \sin(kx)$

Step 2: Inner derivative: $d(kx)/dx = k$

$dy/dx = -Ak \sin(kx)$.

Rough work / extra work

Homework: exponential chain rule

Differentiate $y = 5e^{3x}$.

Solution: Homework: exponential chain rule

Step 1: Keep 5 outside

Step 2: $d(e^{3x})/dx = e^{3x} \cdot 3$

$dy/dx = 15e^{3x}$.

Homework: negative exponent

Differentiate $y = e^{-x}$.

Solution: Homework: negative exponent

Step 1: Outer derivative remains e^{-x}

Step 2: Inner derivative: $d(-x)/dx = -1$

$dy/dx = -e^{-x}$.

Rough work / extra work

Homework: nested sine

Differentiate $y = 3\sin(x^2+2)$.

Solution: Homework: nested sine

Step 1: Keep 3 outside

Step 2: Outer derivative: $\cos(x^2+2)$

Step 3: Inner derivative: $d(x^2+2)/dx = 2x$

$dy/dx = 6x \cos(x^2+2)$.

Higher differentiation lock lines

differentiate again to obtain d^2y/dx^2, d^3y/dx^3, ...

chain rule: outer derivative × inner derivative

S.H.M.: $d^2y/dt^2 = -\omega^2 y$

partial derivative: change one variable, hold the others constant

parametric: $dy/dx = (dy/dt)/(dx/dt)$

Practice

1. Redo all chain-rule questions without seeing the solutions.

2. Write the first four derivatives of sin x and cos x.

3. Prove the S.H.M. differential equation once.

4. Practise one partial and one parametric derivative.

For nested functions, never forget the derivative of the inside.